JM

88%

SOFTWARE CAT 3

MODULE 3,TERM 3

44/50

Instructions

Answer all questions

QUESTIONS

1. Define the term **API** and give one real-life example of its use. (2 marks)
2. State **three** differences between REST and SOAP. (3 marks)
3. Define an **API endpoint** and give one example. (2 marks)
4. Outline **four** main features of Postman. (4 marks)
5. State the HTTP method used to perform each of the following actions: (4 marks)
   (a) View student records
   (b) Add a new student
   (c) Edit a student's details
   (d) Remove a student record
6. State the meaning of each of the following HTTP status codes: (4 marks)
   (a) 201 (b) 401 (c) 404 (d) 500
7. Describe the steps you would follow in Postman to retrieve a list of users from https://jsonplaceholder.typicode.com/users using a GET request. (4 marks)
8. Define **corrective measures** and distinguish them from preventive and detective measures. (3 marks)
9. Name **four** software corrective tools and state the purpose of each. (4 marks)
10. Outline **five** steps followed when performing corrective measures on a faulty computer system. (5 marks)
11. Define **software testing** and state **two** of its objectives. (3 marks)
12. Differentiate between **smoke testing** and **regression testing**. (2 marks)
13. List **four** levels of software testing and briefly describe each. (4 marks)
14. Name the **four** types of software maintenance and give one example of each. (4 marks)
15. Define a **software maintenance schedule** and state **two** of its benefits. (2 marks)

1. Is an application a set of rules that allows different software applications to communicate and exchange data.
Example: A weather application uses a weather service's API to request and display weather information.

2.

REST	SOAP
Can exchange data in different formats eg JSON & XML	Uses XML for its message format
Is lightweight and faster	Is heavier with more built-in security.
Is an architectural style	Is a third protocol

3. Is a specific URL where an API receives requests and returns resources. Example https://jsonplaceholder.typicode.com/users

4. Sending requests.
Collections to organize requests.
Environments and variables
Automated testing and scripts

5. (a) GET (b)
(b) POST
(c) PUT
(d) DELETE

6. (a) created, the resource was successfully created.
(b) Unauthorized, authentication invalid.
(c) Not found
(d) Internal server error

7 Open postman and create a new HTTP request
Select GET from the method dropdown
Enter the URL given in the question https://jsonplaceholder.typicode.com/users
Click send, the expected the response body for the list of users and check the response status.

8 Corrective measures - Action taken to fix a problem after it has occured and restore normal operation.
Preventive measures - Stops problems before they happen.
Detective measures - Identify problems that have occured.

9 Antivirus - removes infection.
Disk repair tools - fix disk errors.
System restore - reverts the system to a working state.
Data recovery software - recovers lost files.

10 Identify and document the problem.
Diagnose the cause
Select a suitable fix
Apply the fix
Test the system and record what was done.

11 Evaluating the software to ensure that it works as exp
Objectives: Find defects and verify that requirements are

12 Smoke testing is a quick check that the main functi
on a new build.
Regression testing re-tests existing features to confir
changes haven't broken them.

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